Homework 12 Solutions

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A reusable clustering skill (`SKILL.md`), applied by a separate AI agent to the Golub leukemia gene-expression data. The one-page report the agent produced is on the next page, and the complete skill is appended after it.

Use of AI tools. I used Claude (Claude Code) to draft `SKILL.md`, this page, and the check code. As the assignment requires, a separate AI agent applied the skill and wrote the one-page report. Every number on this page comes from the code output shown here or from the agent's report, and I checked them against each other.

(a) The skill

`SKILL.md` (4,827 bytes, under the 5,000-byte limit) has YAML `name` and `description` frontmatter and `disable-model-invocation: true`, so it loads only when explicitly invoked. It names no dataset, variables, or results. It requires at least three algorithms (K-means, Ward hierarchical, and spectral clustering) fitted in a common PCA space. It sets the number of clusters and the method from label-free criteria (mean silhouette width and subsample stability, with an admissibility rule and tie-breaks), and it states in bold that the report must fit on one printed page and present clustering results from only the selected method.

(b) Invocation

I saved the skill as `.claude/skills/cluster-select-report/SKILL.md` and applied it with a separate Claude Code agent that started with a fresh context. It wrote only to `homework/week-12/q1b-run/`, a folder holding just a copy of `golub.csv`. The data details were given in the prompt, not in the skill. The agent could still read other files in the repository, including an earlier version of this report, so it was not fully isolated. This is the complete prompt (absolute paths shortened to repository-relative paths):

> Invoke the skill at `.claude/skills/cluster-select-report/SKILL.md`: read it and follow its instructions to analyze the data below. Let the skill guide every analysis and reporting choice.
>
> Data: `homework/week-12/q1b-run/golub.csv`. It has 38 rows, one per biological sample. The feature columns are `gene_1` through `gene_3051` (gene-expression measurements). `sample_id` is an ID column. `leukemia_type` (ALL or AML) is a known label column.
>
> Use R (cluster, mclust and knitr are installed; Quarto 1.10 is available). Work only in `homework/week-12/q1b-run/`. Produce report.qmd and render it to report.pdf, which must be exactly one page. Put the title "Clustering the Golub Leukemia Samples" and the author "Prerith Arunkumar (prerith2)" on the page. Do not git commit anything.
>
> When you finish, reply with: the files you produced, the seed, the preprocessing, the candidate methods and K values, the criteria values for each method's best solution, the selected method and why, the key numbers in the report, the page count of report.pdf, and anything in the skill you found ambiguous or had to decide yourself.

The agent finished in under three minutes without steering. It kept the 500 most variable genes, standardized them, and used the first 10 principal components; it compared the three algorithms for $K = 2, \ldots, 6$ and selected K-means with $K = 3$.

(c) Check

The code below repeats the selected fit independently of the agent's code: same preprocessing, K-means with 50 starts, then the cluster-by-type table, the adjusted Rand index, and the mean silhouette.

```
suppressPackageStartupMessages({library(cluster); library(mclust)})
d <- read.csv("data/golub.csv")
G <- as.matrix(d[, grep("^gene_", names(d))])
top <- order(apply(G, 2, var), decreasing = TRUE)[1:500]
pc <- prcomp(scale(G[, top]))
S <- pc$x[, 1:10]
set.seed(432)
km <- kmeans(S, 3, nstart = 50)
table(cluster = km$cluster, type = d$leukemia_type)
```

```
       type
cluster ALL AML
      1  18   0
      2   1  11
      3   8   0
```

```
c(PCs_var_share = sum(pc$sdev[1:10]^2) / sum(pc$sdev^2),
  ARI = adjustedRandIndex(km$cluster, d$leukemia_type),
  mean_silhouette = mean(silhouette(km$cluster, dist(S))[, 3]))
```

```
  PCs_var_share             ARI mean_silhouette
      0.6764876       0.5077545       0.2608321
```

The check reproduces the report: three clusters of sizes 8, 18 and 12 (cluster numbers are arbitrary). One holds all 11 AML samples and 1 ALL sample, and the other two hold only ALL samples. The 10 components explain 67.6% of the variance, the ARI with the known types is 0.508, and the mean silhouette is 0.261.

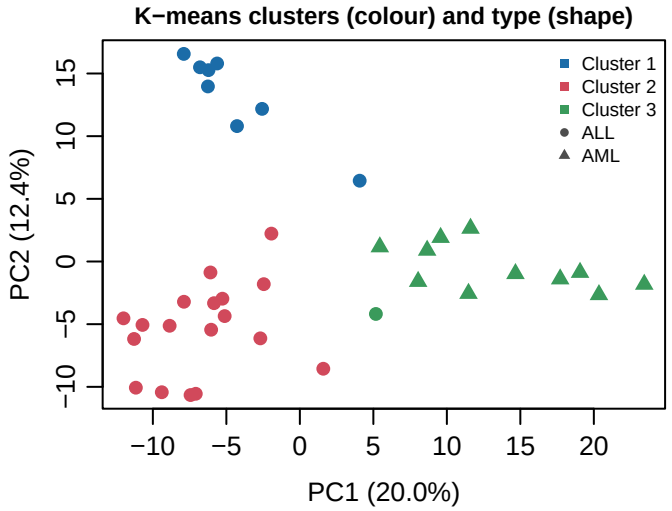
Reviewing the agent's choices, two show where the skill could be tighter. The 70% variance rule and the 10-component cap conflict here (12 components would be needed), and the agent applied the cap, as the report states. The tie-break is worded for two methods, and all three were within 0.02 in silhouette, so the agent applied it across all three and chose the most stable. The interpretation is appropriately cautious: it suggests the split of the ALL samples may reflect B- and T-cell subtypes but says the data contain no subtype labels to confirm it.

Clustering the Golub Leukemia Samples

Prerith Arunkumar (prerith2)

Data. `golub.csv` has n = 38 samples and p = 3051 gene-expression features (p is about 80 times n), 0 missing values, and values centred within each sample; gene SDs range from 0.20 to 1.84. I kept the 500 most variable genes, standardized them, and used the first 10 principal components (67.6% of variance; the 70% rule needed 12, so the 10-PC cap applied) as the common space for all methods. Seed 432; `leukemia_type` was not used until after selection.

Methods and selection. K-means (50 starts), Ward hierarchical clustering (Euclidean), and spectral clustering (Gaussian weights on a symmetric 6-NN graph, median-neighbour bandwidth, normalized Laplacian) were fit for K = 2 to 6. For each I computed the mean silhouette, the mean ARI stability over 50 subsamples of 80%, and the smallest cluster. A solution is admissible if stability ≥ 0.75 and the smallest cluster has ≥ 3 samples; each method's best admissible K maximises silhouette, and methods within 0.02 of the top silhouette are separated by stability. **Selected: K-means, K = 3** (silhouette 0.261, stability 0.942, smallest cluster 8). Comparison: Ward K = 3 (silhouette 0.259, stability 0.934, smallest 7); Spectral K = 3 (silhouette 0.257, stability 0.902, smallest 8). All three are within 0.02 in silhouette, so the higher stability decided.



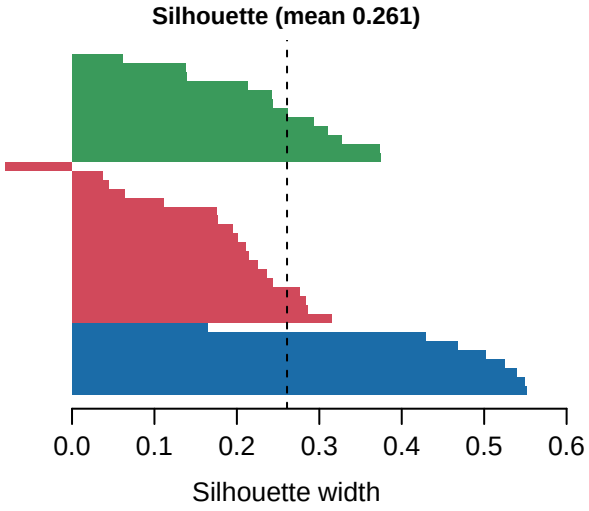


Table 1: Clusters by known leukemia type (ARI = 0.508).

Cluster	ALL	AML	Size
1	8	0	8
2	18	0	18
3	1	11	12

Interpretation. The 3 K-means clusters have sizes 8, 18, 12. Cluster 1 (8 ALL, 0 AML) is marked most by PC2 (SMD 4.53) and gene_1752 (SMD 6.40; 53 genes with |SMD| > 1.5). Cluster 2 (18 ALL, 0 AML) is marked most by PC2 (SMD -1.70) and gene_2065 (SMD 3.20; 50 genes with |SMD| > 1.5). Cluster 3 (1 ALL, 11 AML) is marked most by PC1 (SMD 4.11) and gene_2489 (SMD -3.06; 84 genes with |SMD| > 1.5). So the clusters agree with the known type: one cluster is essentially the AML group and the ALL samples divide in two, which is why the ARI is only 0.508; the ALL split may reflect B- vs T-cell subtypes, but no subtype label was supplied to check this. The structure is stable under subsampling, yet the silhouette of 0.261 means the groups are only moderately separated, so they are best read as real but overlapping expression programmes rather than sharply distinct populations.

Limitations. With n = 38, every estimate is noisy; results depend on the top-500 variance filter, standardization, Euclidean distance and the 10-PC space (67.6% of variance), and the 2-D PC display is not proof of separate populations.

Appendix: SKILL.md

```
---
name: cluster-select-report
description: Compare at least three clustering algorithms on a numeric data set, select one with
label-free criteria, and write a one-page report that presents only the selected method. Use only when
the user explicitly invokes this skill and supplies the data details.
disable-model-invocation: true
---

# Compare clustering methods and write a one-page report

The user supplies: the data file, the feature columns, any ID columns, any known
label columns, the language (R or Python), and the output format. Ask if any is missing.

## Rules

- Use only the feature columns for preprocessing, tuning, the number of clusters,
  and method selection. Known labels may be used only after the final method is
  chosen, to interpret its clusters.
- Fix and report every random seed. Every number in the report must come from
  code output; re-check each quoted number against the output after rendering.
- **The analysis report must fit on one printed page, including all text,
  figures, and tables.**
- **Present clustering results (assignments, cluster sizes, plots, profiles,
  label cross-tables) for only the one selected method.** The other methods may be
  named, with their selection-criterion values in one sentence or one short row
  each, to justify the choice. Do not show their partitions or plots.

## Steps

1. **Inspect.** Report n, p, missing values, and feature scales. Note whether
   p is much larger than n.
2. **Preprocess.** Handle missing values. If p is large (say p > 10n), keep the
   most variable features (for example the top 500) and say how many. Standardize
   features unless they share a meaningful common scale. Compute PCA and keep the
   smallest number of components explaining at least 70% of the variance
   (at most 10). Use these scores as the common space for every method, so the
   comparison does not depend on different inputs. PCA alone is not a clustering
   method.
3. **Candidates.** Fit at least three different algorithms, for each
   K = 2, ..., Kmax with Kmax = min(6, floor(n / 5)):
   - K-means with at least 50 random starts;
   - agglomerative hierarchical clustering with Ward linkage on Euclidean
     distance (average linkage may be added as a variant);
   - spectral clustering: Gaussian similarity on a symmetric k-nearest-neighbor
     graph (k about sqrt(n), bandwidth the median neighbor distance), normalized
     Laplacian, K-means on the row-normalized leading K eigenvectors;
   - optionally PAM or a Gaussian mixture.
4. **Criteria.** For every method and K compute, in the common PCA space:
   - mean silhouette width (Euclidean distance);
   - stability: over B >= 50 random subsamples of 80% of the observations, refit
     and compute the adjusted Rand index (ARI) with the full-data labels on the
     shared observations; report the mean;
   - the smallest cluster size.
5. **Select.** A solution is admissible if mean stability ARI >= 0.75 and the
   smallest cluster has at least max(3, 0.05 n) observations. For each method,
   take the admissible K with the largest silhouette. Choose the method with the
   largest silhouette; if two are within 0.02, prefer the higher stability, then
   the smaller K, then the simpler method (K-means, hierarchical, spectral). If no
   solution is admissible, say so and report the most stable one.
6. **Interpret the selected method.** Refit on all observations. Give cluster
   sizes and the features (or PCs) that differ most between clusters, using
```

```
   standardized mean differences. If labels were supplied, cross-tabulate
   clusters against them and give the ARI. State what the groups appear to
   represent and how confident the evidence allows one to be.
7. **Limitations.** Mention small n, the feature filter and PCA cutoff, the
   dependence on distance and scaling, stability, and that a 2-D display is not
   proof of separate populations.

## One-page report layout

- Title and one or two lines on the data and preprocessing.
- Methods and selection: name the algorithms, state the criteria and the rule, and
  give the selected method's values with one short line of comparison values.
- One figure for the selected method only, at most two panels and about 2.5 to 3
  inches tall: a scatter of the first two PCs colored by cluster (use point shape
  for known labels if supplied), plus a silhouette plot or a profile of the top
  distinguishing features.
- One small table: cluster sizes, or clusters by known label.
- Three to five sentences of interpretation, then one or two of limitations.
- Hide code on the report page; place code and any requested appendix (such as
  this skill) after a page break.
- Render, count the pages of the report section, and if it exceeds one page,
  shorten the text first, then reduce the figure. Re-render until it fits.
```